

Last Discussion Summary

Why local similarity is still Platonic, how each concept maps to project tests, and the flows behind the shared-Hub methodology

Scope. This PDF summarizes the last discussion and includes the final corrected comment. It focuses on the precise version of the Platonic claim: not identical vectors, but different coordinate charts of a partially shared relational manifold.

Running example used throughout. Image: a cat sitting on a chair. Text: "A cat is sitting on a chair."

Semantic item i

image_i: pixels of a cat on a chair

text_i: "A cat is sitting on a chair."

Encoders create different coordinates for the same referent:

DINOv2 image space x_{img_i}

BGE text space x_{bge_i}

GPT-2 text space x_{gpt2_i}

HUB shared coordinates h_i

The question is not: are x_{img_i} and x_{bge_i} numerically the same?

The question is: which relations survive across these spaces?

1. The precise Platonic interpretation

Naive claim: good models converge to the same embedding space. This is too strong. Raw coordinates, dimensions, scales, angles and cosines can be unrelated across encoders.

Precise claim: different models may be different charts of one partially shared relational manifold. A chart is a coordinate system over the same underlying structure. The coordinates can change while some relations survive.

```
Underlying relational manifold M
concept: cat-chair
near: cat-sofa, kitten-chair, pet-on-furniture
far: airplane, spreadsheet, volcano

Encoder A:  $Z_A = f_A(M)$ 
Encoder B:  $Z_B = f_B(M)$ 

Coordinates may differ entirely:
 $f_A(\text{cat-chair}) \neq f_B(\text{cat-chair})$ 

But local relations may survive:
 $N_A^k(\text{cat-chair}) \approx N_B^k(\text{cat-chair})$ 
```

Why local similarity is still Platonic

Local similarity is Platonic in the precise sense because it measures an invariant of the underlying manifold: who is near whom. It does not require the two spaces to share identical axes, metric scale or dimension. If two encoders preserve the same neighbourhood around the same semantic item, they are preserving part of the same relational structure.

This narrows the claim to what can be measured. Local agreement supports a shared relational manifold; it does not prove identical global geometry.

Map analogy

Two maps of Earth can use different projections. Global distances and areas may change, but local adjacency often remains. Spain remains next to France even if the map projection stretches Europe. Likewise, two encoders can warp global geometry while preserving local topology.

2. Concept-to-test assignment

Explained item	Test/status in our project	What it means	Cat-chair example
Local topology / neighbours	PROPOSED NEXT TEST: kNN overlap at $k=\{1,5,10,20,50,100,500\}$	Checks whether the same samples appear near the same item in two spaces.	Does cat-chair have cat-sofa, kitten-chair, pet-on-furniture as neighbours in both image and text spaces?
Distance ranking	PROPOSED NEXT TEST: rank/Spearman over pairwise distances	Checks whether relative order of distances is preserved, even if values differ.	cat-chair should rank closer to cat-sofa than to airplane in both spaces.
Global structure	PERFORMED / EXISTING: CKA-style structural matrix where available	Compares broader Gram-matrix structure; symmetric and dimension-tolerant.	Do all samples have a similar global similarity pattern around the cat-chair item?
Shared linear subspace	PERFORMED / DIAGNOSTIC: Ridge / CCA-style linear mapping where available	Tests whether information can be exported through a linear map.	Can image cat-chair vector be mapped into the text-space direction of the matching caption?

Explained item	Test/status in our project	What it means	Cat-chair example
Near-isometry / same geometry	DIAGNOSTIC / PROPOSED: PCA + Orthogonal Procrustes	Tests whether spaces are almost the same up to rotation/reflection.	After PCA, can a pure rotation align image cat-chair with text cat-chair?
Directional exportability	PERFORMED: R@1/R@5/R@10 retrieval after mapping/head	Tests whether a predicted target vector retrieves the exact paired item.	Does predicted BGE vector for the cat-chair image retrieve its true cat-chair caption/image pair?
AUC verification	PERFORMED / DIAGNOSTIC: pair-vs-random separation	Weaker than retrieval: detects coarse correspondence even if exact top-1 fails.	Is the true cat-chair pair scored above a random dog/airplane pair most of the time?
Hub dimensionality	PERFORMED / METHODOLOGY: SVD/PCA sweep and rows-per-dimension check	Balances retention against statistical support.	Does HUB_512 preserve enough directions for cat-chair discrimination without overfitting?
GPT-2 anisotropy/collapse	DIAGNOSTIC: effective rank, mean cosine, hubness, R@1 vs AUC	Detects loss of pair-specific discrimination.	Many GPT-2 caption vectors become too similar, so cat-chair is not uniquely retrievable.

3. kNN: the local-geometry test

Definition. kNN means k-nearest neighbours. For a sample i , find the k closest other samples inside each space, then compare the two neighbour sets.

```
For sample  $i$  = "A cat is sitting on a chair."  
  
In BGE text space:  
N_BGE^5( $i$ ) = {cat on sofa, kitten on chair, cat indoors, pet on furniture, cat sitting}  
  
In DINO image space:  
N_IMG^5( $i$ ) = {kitten on chair, cat on sofa, dog on chair, cat indoors, cat sitting}  
  
Shared neighbours = 4 out of 5  
kNN-overlap@5 = 4/5 = 0.80
```

Why it is good. kNN is a weak but important Platonic test: it only asks whether the local semantic neighbourhood survived. It is robust to different dimensions and coordinate systems, and it can reveal shared topology even when global metrics or linear maps are weak.

Interpretation. High kNN with low Procrustes means: shared topology, different global geometry. That is not a failure of the Platonic idea; it is the chart version of the Platonic idea.

4. CKA, Ridge, CCA and Procrustes: stronger tests

Test	Strength of claim	What survives if test succeeds	Failure mode
kNN overlap	Weakest, local	Neighbourhood topology	May not imply any global map.
Rank/Spearman	Local-to-global ordering	Relative distance order	Can preserve order but distort metric values.
CKA/RSA	Global structural	Broad similarity fingerprint	Does not prove vectors are exportable.
CCA/Ridge	Linear information	Shared linearly accessible subspace	May miss nonlinear or private information.
Orthogonal Procrustes	Strong geometric	Same geometry up to rotation/reflection	Fails under anisotropy, scaling or nonlinear warp.
R@K after mapping	Operational	Exact paired retrieval after export	Can fail even when coarse correspondence exists.

Increasing strength of evidence:

```
kNN overlap  
  -> distance-rank agreement  
    -> CKA / RSA  
      -> CCA / Ridge  
        -> Orthogonal Procrustes  
          -> R@K exact retrieval
```

Do not collapse these into one statement.
Each test answers a different question about what is shared.

5. Shared-Hub flow: how the project operationalizes charts

The shared-Hub system operationalizes the chart idea. Each encoder speaks its own native coordinate language. The Hub is a learned common coordinate language. A frozen downstream head then tests whether different encoders can use that language consistently.

G1 shared-Hub construction

For each aligned item i:

```
image_i -> DINO-small -> x_small_i (768)
image_i -> DINO-base -> x_base_i (1536)
image_i -> DINO-large -> x_large_i (2048)
text_i -> BGE -> x_bge_i (1024)
text_i -> GPT-2 -> x_gpt2_i (768)
```

Standardize each block, concatenate:

$z_i = [x_{\text{small}} \mid x_{\text{base}} \mid x_{\text{large}} \mid x_{\text{bge}} \mid x_{\text{gpt2}}]$ in $\mathbb{R}^{6144}$

Apply PCA/SVD + whitening:

$Z \rightarrow H$ in $\mathbb{R}^{(8533 \times 512)}$

Each encoder learns its chart-to-Hub map:

$X_k W_k \approx H$

Runtime / transfer flow

New image

```
-> DINOv2-base
-> x_base in  $\mathbb{R}^{1536}$ 
-> W_base
-> h in HUB_512
-> W_head
-> t_hat in BGE_1024
-> cosine retrieval against text/image index
```

Test assignment. This is not only a philosophical diagram. The actual test is whether a head trained on one Hub arm still works when fed Hub vectors produced by another arm. If it works, the arms are not merely similar in wording; they are functionally aligned through the Hub.

6. Directionality: why $A \rightarrow B$ can differ from $B \rightarrow A$

Mapping is directional because one representation may contain information that another lacks, or may expose the shared information more directly. Therefore 7 spaces produce 21 symmetric pairs but 42 directed mappings.

Symmetric structural comparison:

$CKA(A,B) = CKA(B,A)$ -> 21 pairs for 7 spaces

Directional export test:

A -> B may be strong
B -> A may be weaker -> 42 directions for 7 spaces

Example:

DINO-large -> DINO-small may be easier than
DINO-small -> DINO-large

Interpretation:

asymmetry can mean information containment, bottlenecking, or different geometry.

7. GPT-2 collapse flow: why R@1 can fail while AUC survives

AUC and R@1 answer different questions. AUC asks whether true pairs tend to score above random negatives. R@1 asks whether the true partner is the single highest-scoring candidate among many. The second is much stricter.

```
Healthy discrimination
true cat-chair pair similarity    = 0.85
random pair similarity           = 0.20
margin is large -> AUC good, R@1 likely good

Collapsed / anisotropic geometry
true cat-chair pair similarity    = 0.92
random pair similarity           = 0.88
a few wrong candidates           = 0.93, 0.91

Result
AUC can remain good because true > typical random.
R@1 can fail because a few negatives outrank the true pair.
```

Possible causal chain

```
GPT-2 intrinsic anisotropy
-> low effective variation
    -> restrictive Hub / low-rank projection
        -> additional loss of discrimination
            -> GPT-2 arm collapse
                -> R@1 crashes
                    -> AUC may remain respectable
```

Methodological interpretation. A collapse in R@1 does not by itself prove there is no shared structure. It may mean the pipeline no longer resolves that structure at pair-specific retrieval precision.

8. Corrected final comment: the exact wording

The following is the final corrected formulation from the last discussion, with the Gromov-Wasserstein point made precise.

Why this is still Platonic - the precise version

The naive claim that sufficiently capable models converge to an identical embedding space is too strong. The measurements instead support a narrower interpretation: different models can behave as different coordinate charts of a partially shared relational manifold. Their coordinates, dimensionalities, and global metrics may differ substantially, while some relational invariants - especially local neighborhood structure - survive across representations. Linear mappings show that part of this shared structure is globally linearly accessible, while their failure to reach the full target ceiling shows that the spaces are not fully equivalent. Gromov-Wasserstein further demonstrates the boundary of geometry-only alignment: relational shape was sufficient for coarse distribution-level alignment but not for recovering exact instance correspondences. Under this interpretation, the Platonic hypothesis becomes a testable claim about which invariants survive across models - neighborhood topology, rankings, global structure, linear subspaces, or near-isometry - rather than a claim that all models converge to identical vectors.

9. Final evidence flow

Evidence flow

1. Aligned semantic items
image: cat on chair
text: "A cat is sitting on a chair."
2. Multiple encoders create different coordinate charts
DINO / BGE / GPT-2 / BERT / SBERT / others
3. Test which invariants survive
kNN -> local topology
ranks -> relational ordering
CKA -> global structural fingerprint
CCA/Ridge -> shared linear information
Procrustes -> near-isometric geometry
R@K -> exact operational export
4. Build or evaluate a shared Hub
encoder -> W_encoder -> HUB_512 -> W_head -> BGE_1024 -> retrieval
5. Interpret failures carefully
R@1 collapse does not automatically mean no shared structure.
It may mean insufficient pair-specific resolution.
6. Precise Platonic conclusion
Not: all models learn the same vector space.
Yes: models may preserve some invariants of a shared relational manifold.

10. One-page takeaway

Local is good because it is the most robust invariant. If different encoders disagree globally but agree on local neighbours, they still share part of the same semantic topology.

Linear maps are stronger evidence, not the only evidence. A successful Ridge or Procrustes result means more of the shared structure is globally accessible. But a weak linear map does not erase local Platonic evidence.

The Hub is a functional chart test. If different encoders can enter HUB_512 and use the same frozen head, they are speaking a partially shared coordinate language.

The careful claim is narrow and testable. The project should not claim identical spaces. It should claim measured survival of specific invariants: neighbourhood, ranking, global structure, linear subspace, near-isometry and retrieval exportability.

Recommended wording for report

Different encoders should be treated as different charts over a partially shared relational manifold. The empirical task is to identify which invariants survive across charts. Local neighbourhood agreement is therefore not a weak philosophical fallback; it is the first measurable Platonic invariant. Stronger tests - CKA, CCA, Ridge, Procrustes and R@K - progressively ask whether that shared structure is global, linear, near-isometric or operationally exportable.

Appendix: Manifold, Ambient Space, and Charts

This appendix adds the missing geometric distinction behind the project: ambient vector space, manifold, and chart. It explains why local similarity is not a weak consolation prize, but often the most robust evidence that two models preserve the same underlying relational object.

1. What “manifold” means here

In this discussion, manifold is the right word. A manifold is a space that can be complicated globally but looks simple and Euclidean when viewed locally. A sheet of paper rolled into a cylinder is curved in 3D globally, but a tiny patch on it still looks like a flat 2D plane.

globally complicated geometry + locally simple geometry

This is exactly the pattern we care about in embeddings: a model can place vectors in a large high-dimensional ambient space, while the meaningful semantic data occupies a smaller, structured, locally coherent region inside it.

2. Ambient space vs intrinsic structure

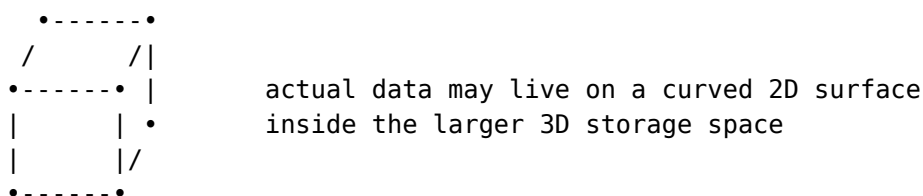
If an embedding model outputs vectors in $\mathbb{R}^{1536}$, every item is stored using 1536 coordinates. But those 1536 coordinates are the ambient storage space, not necessarily the true number of semantic degrees of freedom. The actual semantic structure may behave like a much smaller manifold embedded inside that large space.

x in $\mathbb{R}^{1536}$

actual semantic structure $\approx$ 20-dimensional manifold inside $\mathbb{R}^{1536}$

Ambient space	Manifold	Meaning
$\mathbb{R}^{1536}$ coordinates	low-dimensional curved semantic surface	where useful relations live
where vectors are stored	how data is intrinsically organized	what tests should try to recover

Ambient 3D analogy:



3. Direct connection to the project

Your aligned item can be represented by different modalities and encoders. For the same semantic item:

Image: a cat sitting on a chair.
Text: "A cat is sitting on a chair."

One model may produce x_A in $\mathbb{R}^{768}$, while another produces x_B in $\mathbb{R}^{1536}$. Raw cosine between those coordinates may be meaningless, especially across different spaces. The useful question is therefore not whether x_A numerically resembles x_B , but whether the local neighborhood around the same item is preserved.

x_A in $\mathbb{R}^{768}$
 x_B in $\mathbb{R}^{1536}$
raw cosine(x_A , x_B) may tell us essentially nothing

Space A neighborhood	Space B neighborhood
$N_A(x) = \{a, b, c, d\}$	$N_B(x) = \{a, c, b, d\}$
same local semantic region	same local semantic region, different order

The coordinates changed, but the local relational structure survived.

4. Charts: why coordinates do not need to match

A chart is a local coordinate system on a manifold. One model may describe a region using coordinates ($u_1, u_2, \dots, u_{768}$), while another model describes the same region using ($v_1, v_2, \dots, v_{1536}$). The coordinates do not need to match. What should survive is some invariant structure.

Model A chart: ($u_1, u_2, \dots, u_{768}$)
Model B chart: ($v_1, v_2, \dots, v_{1536}$)

Should not be required	Should be tested
same raw vector values	who is near whom
same global coordinate system	relative distances / rankings
same dimension	local geometry and topology

This motivates the precise phrase used in the project:
different charts of a shared relational manifold

5. Why k-NN agreement is important

k-NN agreement tests whether models preserve the same local manifold neighborhoods, even when their coordinate systems are radically different. This is why local similarity is not a minor result: for a manifold, local neighborhoods are often the most reliable invariant across charts.

$$N_A^k(i) \approx N_B^k(i)$$

Example:
 $N_A(\text{cat-chair}) = \{\text{cat-sofa}, \text{kitten-chair}, \text{pet-furniture}, \text{indoor-cat}\}$
 $N_B(\text{cat-chair}) = \{\text{kitten-chair}, \text{cat-sofa}, \text{indoor-cat}, \text{pet-furniture}\}$

If the overlap is high, the models agree on the local semantic neighborhood of the item. They may disagree on coordinates, distances, or angles, but still preserve part of the same underlying relational manifold.

6. Why this remains Platonic, but in the defensible form

The naive Platonic claim would be: all good models converge to the same embedding space. That is too strong. The defensible claim is narrower: different models may construct different coordinate representations of a partially shared underlying relational manifold.

Same vectors? No
Same global coordinate system? No
Same local relational geometry? Often yes

So the project should not claim that all models learn identical spaces. It should claim that some invariants of the underlying relational structure can survive across encoders: neighborhoods, distance rankings, topology, shared subspaces, or sometimes even a near-isometric geometry.

Naive claim	Precise claim
All models converge to the same embedding space.	Different models may be different charts of a shared relational manifold.
Rejected by coordinate mismatch and weak raw cosine.	Tested by kNN, rank agreement, CKA, CCA/Ridge, Procrustes, and retrieval.

7. Final mental model

semantic item

- > image/text modalities
- > different encoders
- > different ambient vector spaces
- > different charts
- > test which invariants survive

Most important distinction:
ambient space != manifold != chart

That distinction - ambient space, manifold, and chart - is the most important conceptual tool for interpreting the results. It explains why local similarity can be meaningful, why raw cosine can fail, why linear maps can work partially, and why the project should speak about shared relational structure rather than identical embedding spaces.